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THE HONG KONG POLYTECHNIC UNIVERSITY

DEPARTMENT OF COMPUTING

EXAMINATION

Subject : COMP4423 Computer Vision

Session : 2023 / 2024 Semester II

Date : 8 May 2024 Time : 12:30 - 14:30

Time Allowed: 2 Hours

Subject Examiner(s): Prof. WEI Xiaoyong

This question paper has a total of 5 pages.
(Some pages may be intentionally omitted.)

Instructions to Candidates:

The exam is an open book exam.
Students are allowed to bring 5 pages of A4 size material to the exam.
Electronic devices are not allowed.
Students should write their answers on the answer sheet. Only the answers on the answer sheet will be graded.

Do not turn over the page until you are told to do so!

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Section C: Fill-In-Blanket Questions (20 marks)

21. _____ is a mechanism used in deep learning neural networks to selectively focus on certain parts of an input, improving the model's ability to learn from the data. _____ is a technique used in computer vision to visualize which parts of an image contributed the most to a particular classification decision made by a deep neural network.

22. Gaussian blur and median filtering are two types of image processing techniques used in computer vision for _____. _____ (Gaussian blur/Median filtering) is a better choice for preserving the details of an image.

23. _____ (Softmax, Kernel, Max-Min, Gaussian) is a mathematical function used in neural networks to introduce non-linearity into the network. _____ is a type of neural network layer that is commonly used in image processing tasks to extract features from an input image.

24. _____ is the process of updating model parameters to minimize the loss function using the gradient of the loss function. The _____ is a hyperparameter that controls the step size of the process and can affect the speed of convergence and the stability of the algorithm.

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Section D: Short-Answer Questions (30 marks)

25. You are very fond of your dogs and cats and want to know what they do when nobody is home. You have set up several surveillance cameras, but your boss does not allow you to check them frequently. Consequently, you decide to develop algorithms to assist you. To explain your approach, please provide answers to the following questions. Please keep your responses within 10 words unless otherwise specified.



Assume you have only one cat, and your open window often allows the cat to sneak out. The first thing you want to determine is whether your cat is at home.

1) Can you utilize any pretrained object detection model for this task? Name one of them. (2 points)
2) How do you ascertain whether your cat is at home or playing outside at a specific time? (50 words, 5 points)
3) If you have multiple cats, what additional steps must you take alongside the existing pretrained model? (20 words, 5 points)
4) In the scenario with multiple cats, can you employ traditional features when training the new model? Name a situation and at least one feature. (30 words, 4 points) What are the advantages and disadvantages of traditional features compared to deep features? Provide one for each. (4 points)

Your friend Tom is interested in your algorithms and wants to apply them to his cats.

5) Which model do you need to retrain for Tom's cats? (2 points)
6) Why do you have to retrain it? (3 points)
7) Tom discovered that the algorithms work well during the day but not at night. What is the most likely reason? (2 points)
8) How can you resolve the issue during the night? (3 points)

** END **